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Topic Paper - 2021

Submitted to
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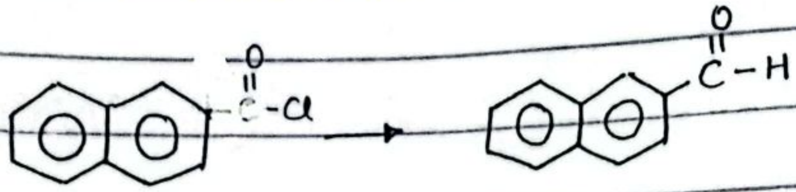
Short Answers

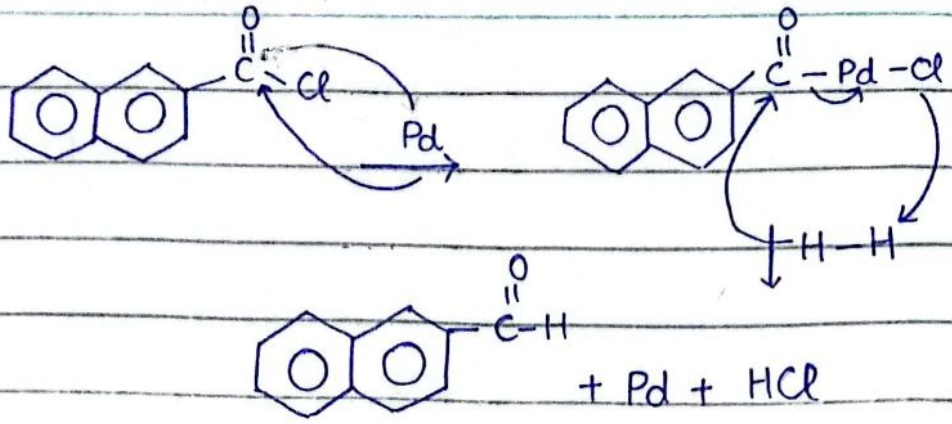
(ii)

Conversion of the acid chloride into aldehyde



Aldehydes can be prepared by the reduction of acid chlorides. Acid chloride being more reactive than product aldehyde it is possible to selectively reduce acid chlorides. One procedure for accomplishing this selective reduction is catalytic hydrogenation. The method is called **Rosenmund Reduction**.

In this method, acid chloride is hydrogenated in presence of a catalyst such as palladium deposited on barium sulphate and poisned with lead or sulphur.



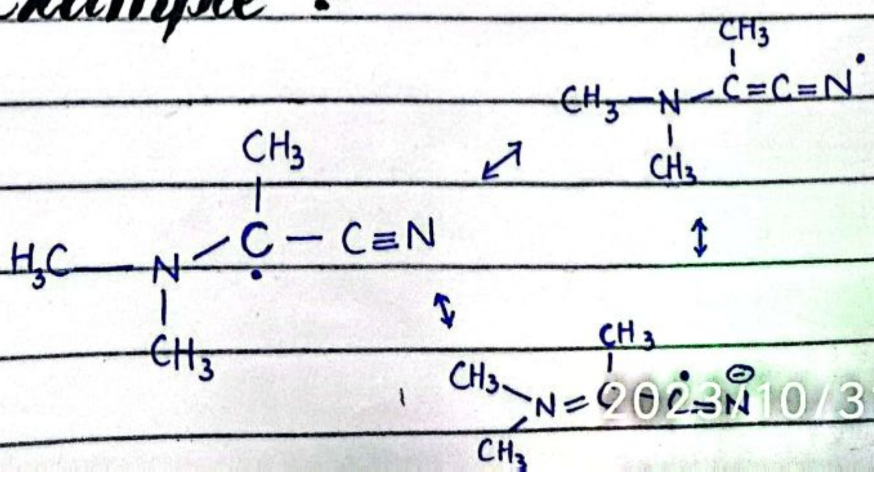
Captodative effect

The captodative effect is the stabilization of radical by a synergistic effect of an electron-withdrawing substituent and an electron-donating substituent.

The electron-withdrawing group (EWG) is sometime called captor group.

The electron-donating group (EDG) is dative substituent.

For Example :-



In this example; radical is delocalized between the captor nitrile (−CN) and the dative secondry amine ($-N(CH_3)_2$) thus stabilizing the radical center.

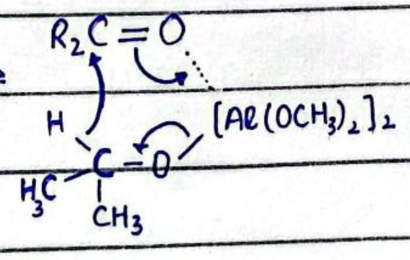
— (iv) —

Meerwein - Ponndrof Verley Reagent use

MPV reduction is a chemical reaction used to reduce ketones and aldehyde to their corresponding alcohols, with aluminium alkoxide.

Mechanism :

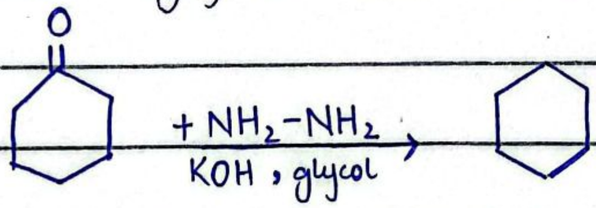
$$R_2C=O + Al[OCH(CH_3)_2]_3 \rightleftharpoons$$



$$R_2CH-OH + Al[OCH(CH_3)_2]_3 \xrightleftharpoons{(CH_3)_3C-OH} R_2CHO\,Al\,[OCH(CH_3)_2]_2 + (CH_3)_2CO$$

Wolf-Kishner Reduction

It is useful method for direct conversion of carbonyl group to a methylene $-CH_2-$ group. The reagent used is hydrazine and KOH in ethylene glycol or N,N-Dimethyl formamide.



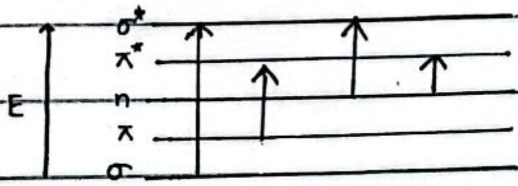
Clemmensen Reduction

It is a direct reduction of a carbonyl group to a methylene group. It involves refluxing the aldehyde or ketone with amalgamated zinc and HCl.

$$C_6H_5\overset{O}{\overset{\|}{C}}-CH_3 \xrightarrow[HCl]{Zn(Hg)} C_6H_5CH_2-CH_3$$

Electronic Transitions in UV spectroscopy

The following electronic transition are involved in ultravoilet and visible region :



$\sigma \rightarrow \sigma^*$ and $n \rightarrow \sigma^*$

$\pi \rightarrow \pi^*$ and $n \rightarrow \pi^*$

$\sigma \rightarrow \sigma^*$ occur in Compounds in which all valence electrons are involved in single bond formation such as saaturated hydrocarbon. e.g propane shows λ_{max} at 135 nm.

$n \rightarrow \sigma^*$ occur in Compounds containing non-bonding electrons on oxygen or heteroatoms. for example CH_3OH shows λ_{max} at 183 nm.

$\pi \rightarrow \pi^*$ occurs in Compound with unsaturated centers ; e.g ethylene shows λ_{max} at 171nm

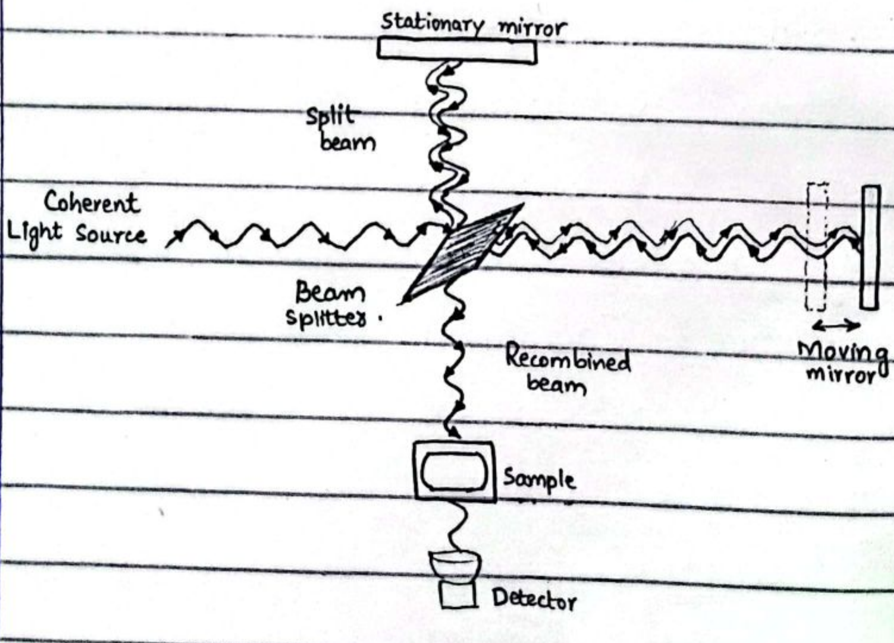
$n \rightarrow \pi^*$ occur in Compounds having heteroatoms and involving double bond ; e.g around

Principle of FTIR

FTIR stands for Fourier transform infrared. It is most common technique of infrared spectroscopy.

The technique based on principle that when ifrared radiation (IR) passes through sample, some of the radiation is absorbed. The radiation that passes through the sample is recorded.

An FTIR spectrometer simultaneously collects high-resolution spectral data over a wide spectral range.



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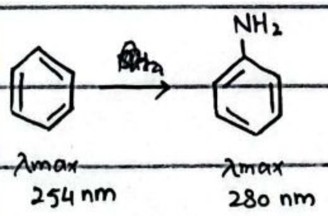
Red Shift

When the absorption maximum shifts toward longer wave-length then it is called red shift.

It is also known as bathochromic shift.

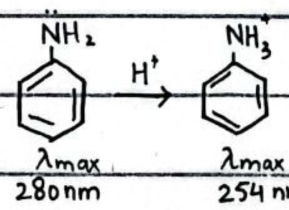


In this, conjugation increases, so λ_{max} increases.

Blue Shift

When the absorption maximum shifts toward shorter wave-length then it is called blue shift.

It is also known as hypsochromic shift.



In this, conjugation decreases, so λ_{max} decreases.

Long Answers

Q No. 1

Applications of UV-visible spectroscopy

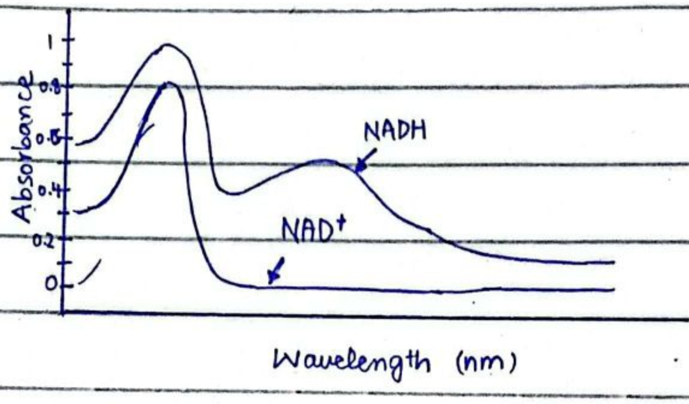
The UV-visible has innumerable applications :

- Chemical kinetics
- Determination of pKa value
- Estimation of nucleotide
- Detection of impurities
- Structural eludication of organic compound.
- Identification of geometric isomers

Chemical Kinetics

The UV-spectroscopy is used in determining rate constants, equilibrium constant, acid-base dissociation constant etc for chemical reaction. The rate of any reaction can be measured long as one of the reactants or one of the products absorb UV at a wavelength at which the other reactants and product have little or no any absorbance.

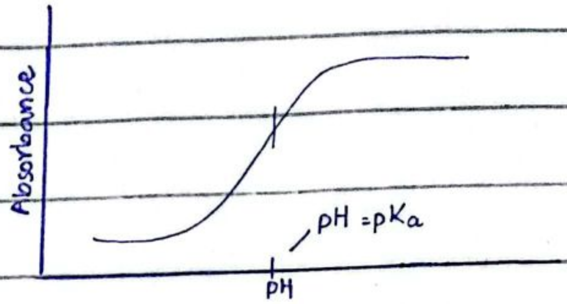
Formation of NADH from NAD^+ in enzymatic reaction.



Determination of pKa

The pKa of a compound can be determined by UV spectroscopy if either the acidic form or the basic form of the compound absorbs UV.

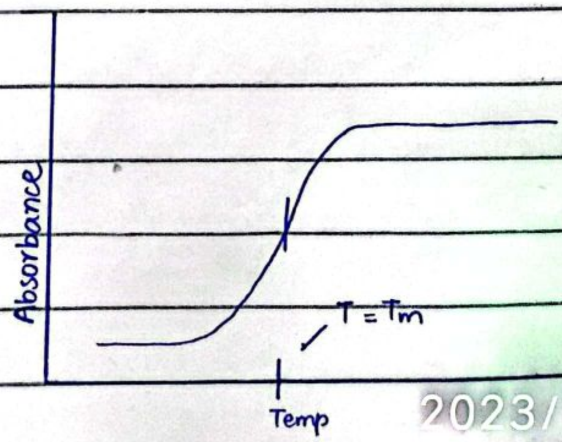
The phenoxide ion has λ_{max} at 287nm. If the absorbance at 287 nm is determined as a function of pH, the pKa of phenol can be ascertained by determining the pH at which exactly one-half the increase in absorbance has occured.



Estimation of Nucleotide

UV-spectroscopy can also be used to estimate the nucleotide composition of DNA. The two strands of DNA are held together by both A-T base pairs and G-C base pairs.

For double stranded DNA Tm (melting temperature of DNA) increases with increasing number of G-C base pair because they are held together by three hydrogen bonds. Therefore; Tm can be used to estimate G-C base pairs.



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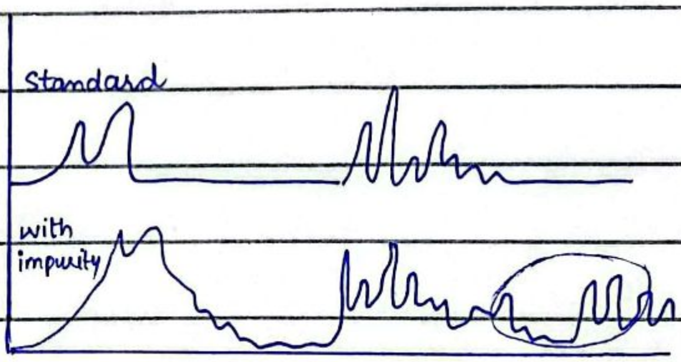
Detection of impurities

UV absorption spectroscopy is one of best method for determination of impurities in organic molecules.

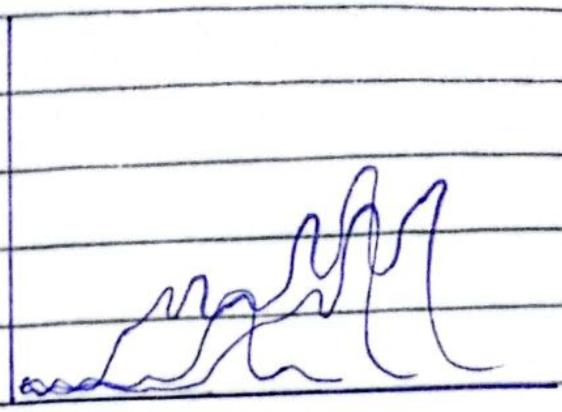
Additional peaks can be observed in UV spectra of substances due to presence of impurities in impure sample and it can be compared with spectra of standard raw material.



Structure eludication of organic compound

UV-spectroscopy is useful in the structure eludication of organic compound or molecules, The presence or absence of unsaturation, the presence of heteroatoms.

This technique is used to detect the presence or absence of functional group



UV spectra showing presence of unsaturation

Identification of Geometric isomers

Cis and trans form can be identified by using UV spectro-scopy.

Cis form absorb UV rays at the shorter wavelength.

Trans form absorb UV rays at the longer wavelength.

Q No. 1
(b)

Applications of Synthetic Free Radicals

Free radicals can be use in synthesis of many industrially important compounds.

Polymerization
Dimerization
Synthesis of chloroform
Synthesis of aldosteron 21- acetate
Synthesis of large ring
Synthesis of dyes and drugs.
Synthesis of diyne.

Polymerization

Free radical polymerization is important reaction for synthesis of many useful polymer like polystyrene, polyvinyl chloride, polyethylene etc.

$$R-\overset{\overset{O}{\|}}{C}-O-O-\overset{\overset{O}{\|}}{C}-R \rightarrow 2R-\overset{\overset{O}{\|}}{C}-O^{\bullet} \rightarrow 2R^{\bullet} + 2CO_2$$

$$R^{\bullet} + CH_2-\underset{\underset{Cl}{|}}{CH} \longrightarrow R-CH_2-\dot{C}H-Cl$$

$$R-CH_2-\underset{\underset{Cl}{|}}{\dot{C}H} + CH_2=\underset{\underset{Cl}{|}}{CH} \rightarrow R-CH_2-\underset{\underset{Cl}{|}}{CH}-CH_2-\underset{\underset{Cl}{|}}{\dot{C}H}$$

$$\left[\underset{\underset{Cl}{|}}{CH}-CH_2-\underset{\underset{Cl}{|}}{CH} \right]_n$$

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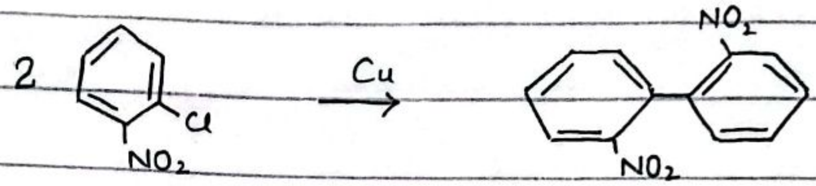
Dimerization

Symmetrical biphenyls may be prepared by heating aryl halides with copper powder.

For example : 2,2'-dinitrobiphenyl



Chloroform Synthesis

The chlorination of methane, gives a mixture of methyl chloride, dichloromethane, chloroform and carbon tetrachloride.

These reactions are of great importance industrially because of these products can be separated by efficient fractional distillation.

$$CH_4 + Cl-Cl \rightarrow CH_3Cl + HCl \text{ (chloromethane)}$$
$$CH_3Cl + Cl-Cl \rightarrow CH_2Cl_2 + HCl \text{ (dichloromethane)}$$
$$CH_2Cl_2 + Cl-Cl \rightarrow CHCl_3 + HCl \text{ (chloroform)}$$
$$CHCl_3 + Cl-Cl \rightarrow CCl_4 + HCl \text{ (tetrachloromethane)}$$

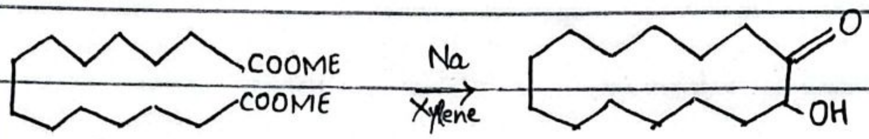
Synthesis of aldosteron 21-acetate

The photolysis of nitrites use in the

Synthesis of aldosteron 21-acetate from corticosterone acetate.

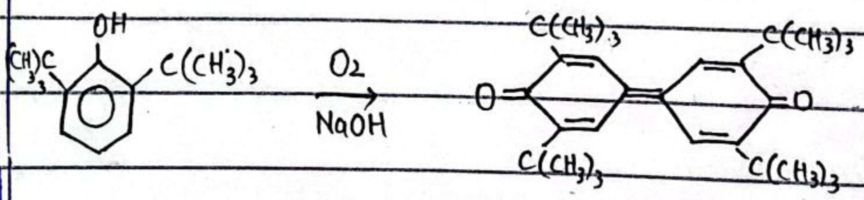
Synthesis of large ring:

For synthesis of large ring from medium, the acyloin condensation is one of the best method.



Synthesis of drugs and dyes:

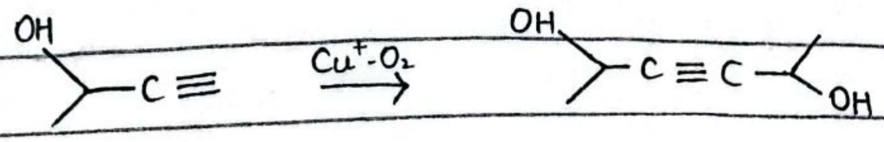
Oxidative coupling of phenol are use for synthesis of some dyes and drugs.



2,6 ditertiarybutyl phenol   biphenyl quinone (red brown) 98%.

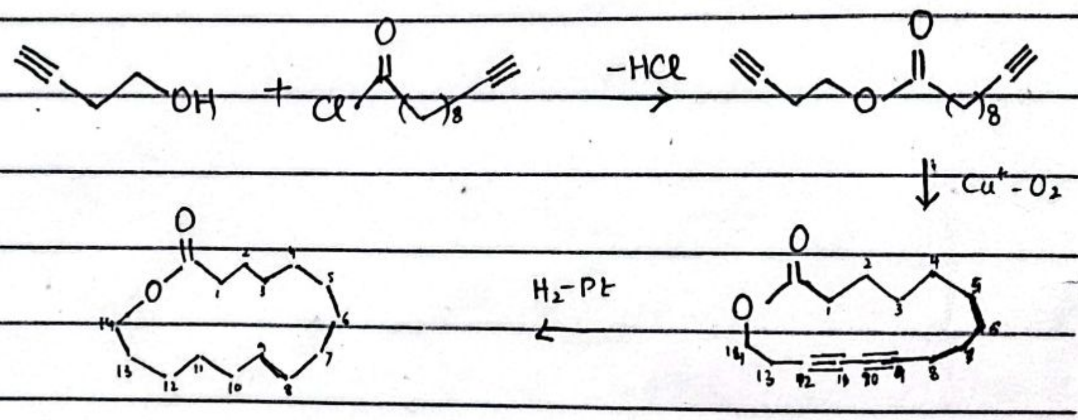
Synthesis of diyne:

The coupling reaction on 3-hydroxy-1-butyne gives a diyne used in synthesis of β-carotene.



Synthesis of cyclic compound

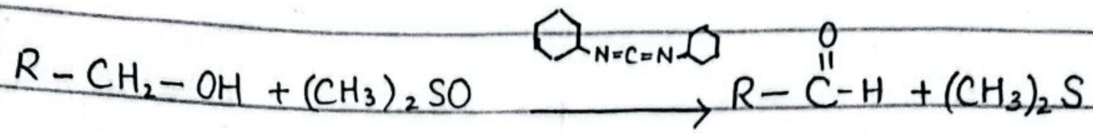
Terminal diynes can couple both inter-molecularly. The macrocyclin lactone which is partly responsible for sweet odour of angelica roots and is used in scents, has been synthesized in high yeild by this mean.



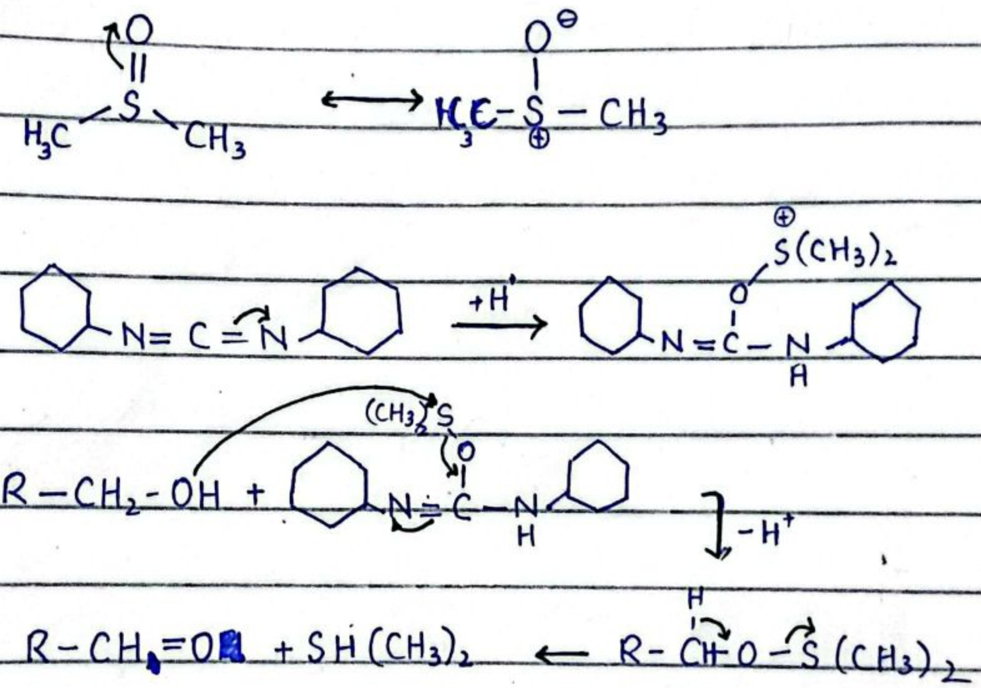
Q No. 2

Corey Kim Oxidation

It is the oxidation reaction used to synthesize aldehyde and ketones from primary and secondry alcohol.

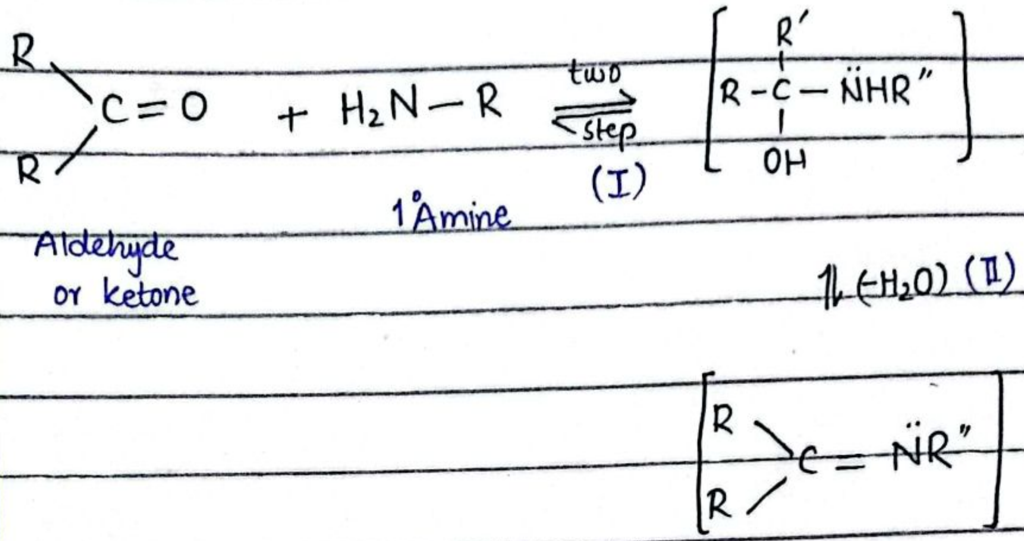


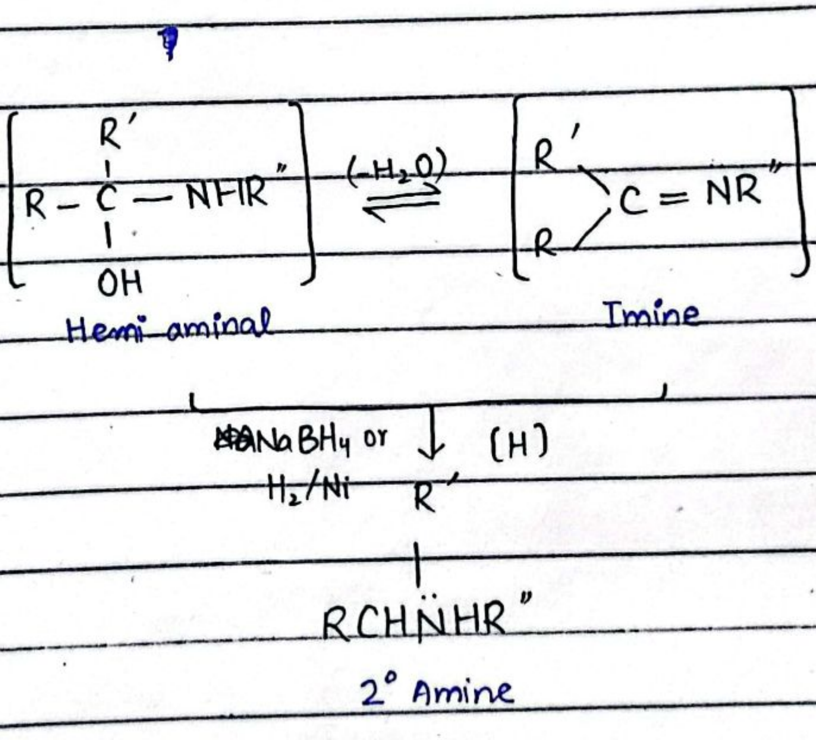
Mechanism :



Reductive Amination

The reactions in which aldehydes or ketones are reduced as well as aminated (converted into amines) are known as reductive amination.



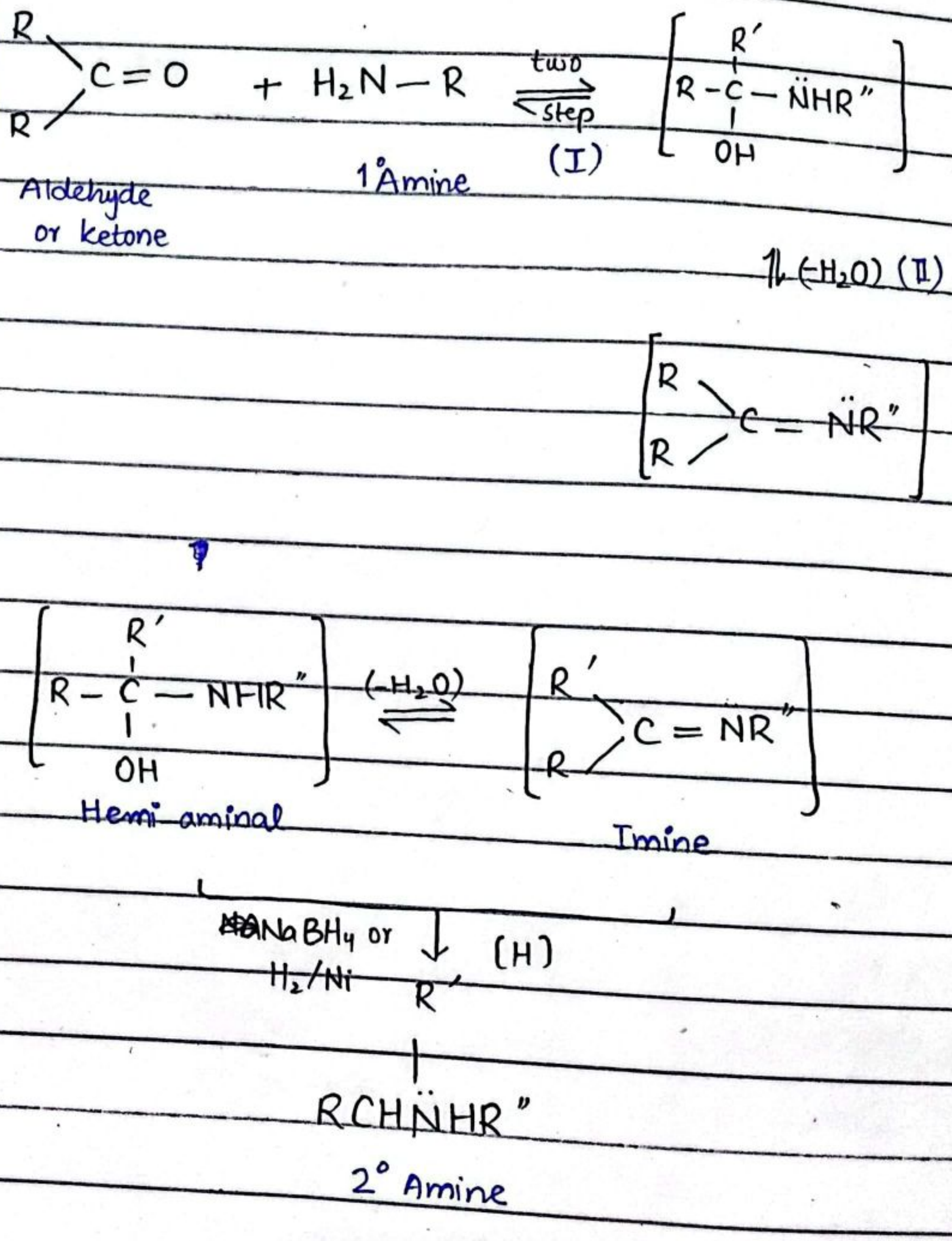


In above mechanism there are two possible pathways to product.

Via an aminoalcohol (hemiaminal)

Via an imine

General Mechanism



In above mechanism there are two possible pathways to product.

- Via an aminoalcohol (hemiaminal)
- Via an imine

These reaction differ from Hoffmann alkylation of ammonia in that the formers can be selectively used to prepare pri-, sec-, or t-amines whereas latter gives a mixture of amines.

Methods :

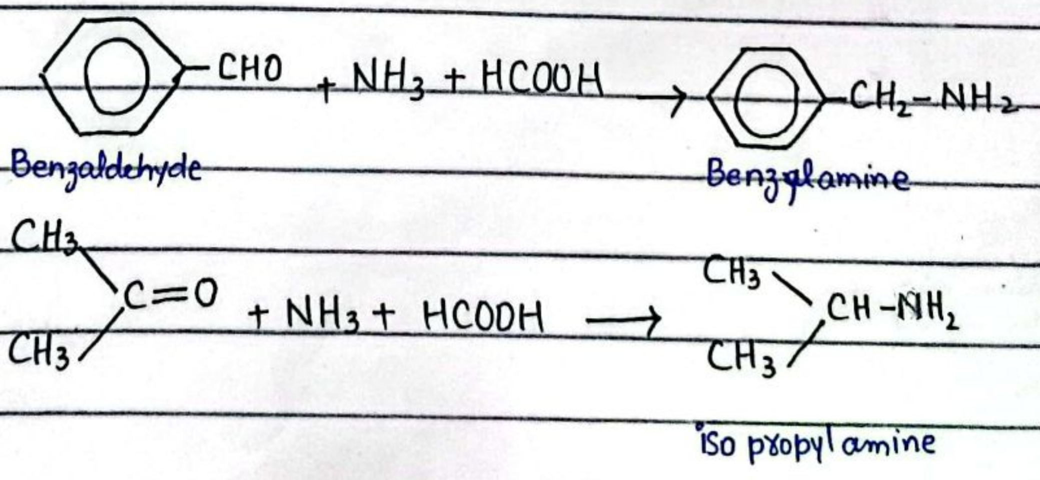
Leukart Method

Wallach method

Eschweiler - Clark method.

Leukart Method :-

According to this method pri- amines are obtained when aldehydes and the ketones are treated with ~~a~~ mixture of NH_3 and HCOOH.



Wallach Method

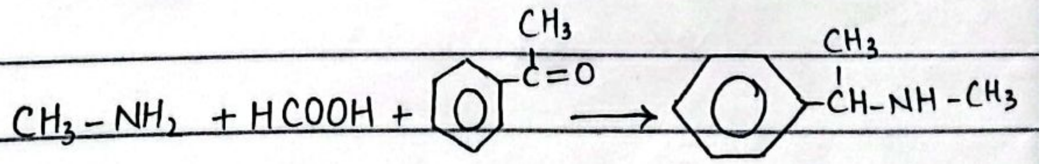
This method can be used to prepare pri, sec, t-amine.

This is also the modification of Leukart method.

In this, in addition to NH_3, pri or sec-amine may also be used. Thus ammonia gives pri-, pri-amine gives sec-amine and a sec-amine gives a ter. amine.

$$NH_3 + HCOOH + \underset{C_2H_5}{\overset{C_2H_5}{>}}C=O \longrightarrow \underset{C_2H_5}{\overset{C_2H_5}{>}}CH-NH_2$$

pri-amine



sec-amine

$$\underset{CH_3}{\overset{CH_3}{>}}NH + HCOOH + CH_3CHO \longrightarrow CH_3CH_2-N\underset{CH_3}{\overset{CH_3}{<}}$$

t-amine

Eshweiler-Clark Method :

This method is used to change pri-, sec- amines to t-amines with a methyl substitution. Primary or sec. amine are treated with a mixture formaldehyde and formic acid, to get N-methyl or N,N-dimethyl amine.

$$CH_3-NH_2 \xrightarrow{CH_2O + HCOOH} CH_3-N\begin{smallmatrix} CH_3 \\ CH_3 \end{smallmatrix}$$

p-amine — t-amine

Q No. 3

(b)

Mechanism & Reaction

$$A \xleftarrow{LiAlH_4} H_3C-CH \equiv CH-\overset{\overset{O}{\|}}{C}-H \xrightarrow{Al(O\text{-}Pr)_3} B$$

$$CH_3-CH=CH-\overset{\overset{O}{\|}}{C}-H \longrightarrow B$$

1- $H_3C-\overset{H}{\overset{|}{C}}=CH-\overset{\overset{O}{\|}}{C}-H + Al[OCH(CH_3)_2]_3 \longrightarrow$

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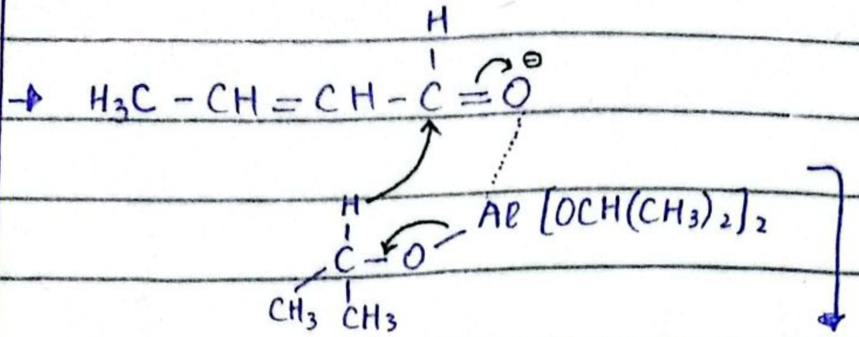


$CH_3-CH=CH-CH_2-O-Al[OCH(CH_3)_2]_2$

$+(CH_3)_2CO$

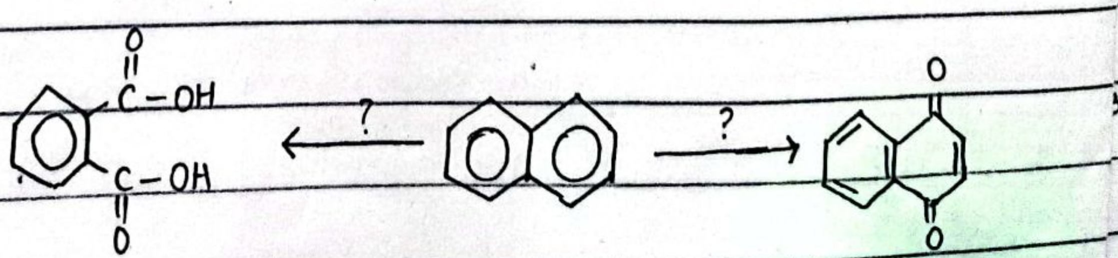
2 - $(CH_3)_3-C-OH + O(CH_2-CH=CH-CH_3)(Al[OCH(CH_3)_2]_2) \rightarrow CH_3-CH=CH-CH_2-OH$

$H_3C-CH=CH-\overset{O}{\overset{\|}{C}}-H \longrightarrow A$

$H_3C-CH=CH-\overset{O}{\overset{\|}{C}}-H + H-AlH_3 + Li^+ \rightarrow CH_3-CH=CH-\underset{H}{\overset{\ddot{O}-AlH_3Li^{\oplus}}{C}}$

$\xrightarrow{H^+} H_3C-\underset{H}{\overset{OH}{C}}-H$

dax



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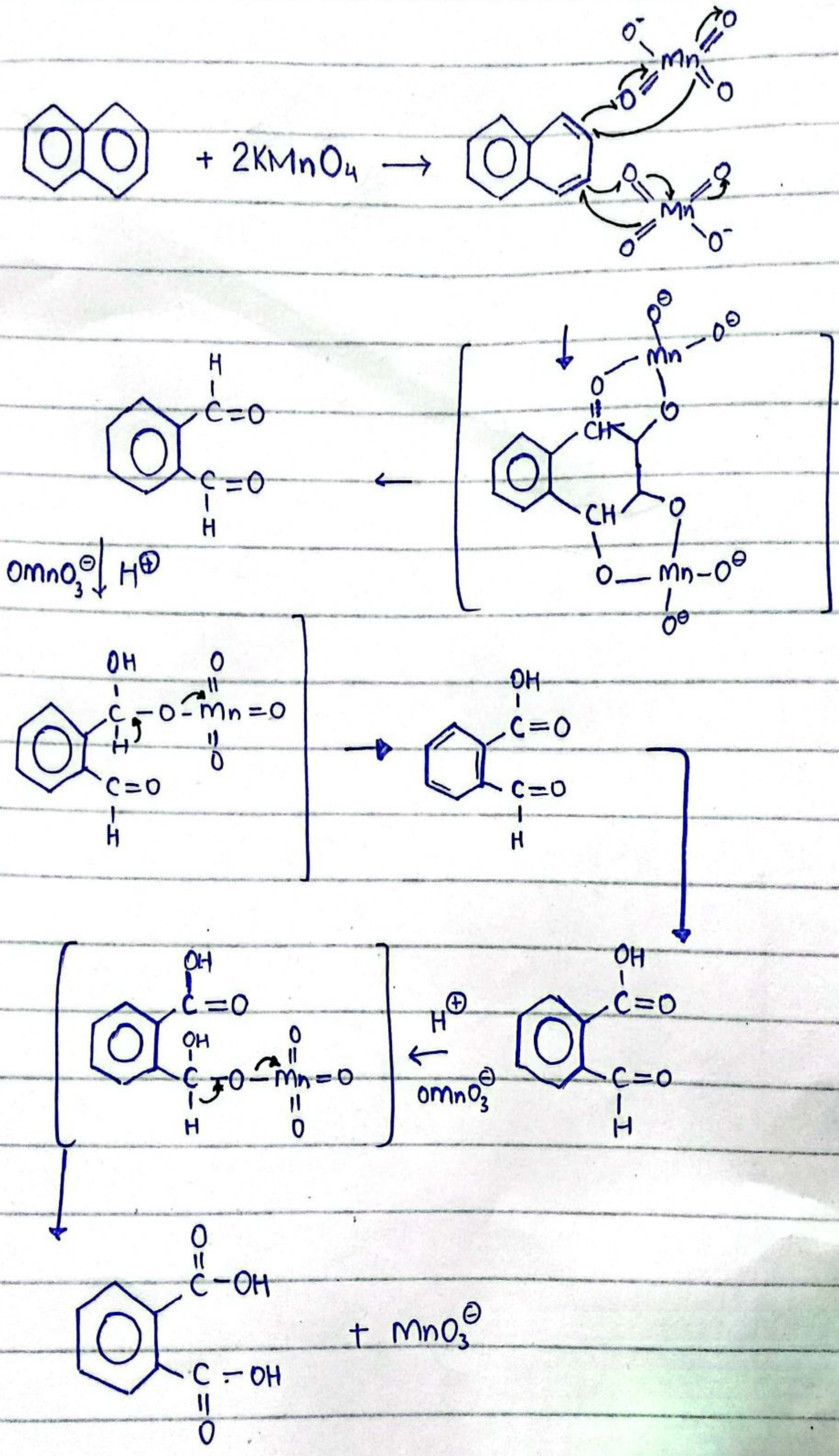

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